Project Summary

PharmaAI: Intelligent Medicine Verification & Safety Network

1. Problem Statement

The modern pharmacy is a complex landscape where two major threats endanger patient lives: Counterfeit Drugs and Adverse Drug-Drug Interactions (DDIs).

The Challenge: Counterfeit medications often look identical to genuine ones, but lack critical regulatory data like batch numbers, proper manufacturing dates, or verified clinical ingredients. For a common patient, there is no way to verify if the medicine in their hand is legitimate.

The Gap: Most existing solutions are "siloed." A user might find an app to check interactions, but it uses static, outdated databases. They might find a separate tool for OCR, but it doesn't cross-reference with live FDA or RxNorm records.

Significance: Medication errors and "look-alike" drugs cause thousands of preventable hospitalizations. There is an urgent need for a unified system that verifies a drug's "identity card" and its safety profile simultaneously.

2. Architecture & Data Flow

Our system follows a modular pipeline architecture that transforms raw visual data into actionable medical intelligence.

Input Stage: The system accepts either a high-resolution image of a medicine strip/bottle or a manually entered drug name.

Vision Processing: Using a Two-Pass OCR engine, the system extracts key pharmaceutical entities: Brand Name, Generic Composition, Batch Number, Manufacturing Date, and Expiry Date.

Identity Resolution: The extracted text is sent to the Drug Resolver. This module uses the RxNorm (National Library of Medicine) API to find the canonical RxCUI (unique ID) and PubChem to fetch the molecular formula and chemical properties.

Safety & Risk Engine: The resolver passes the canonical drug name to the OpenFDA Engine, which scrapes real-time manufacturer labels for "Boxed Warnings," "Contraindications," and "Drug Interactions."

Authenticity Scoring: A logic-based module evaluates the "completeness" of the label. If critical fields like the Batch No. are missing or if the manufacturer cannot be verified against FDA/CDSCO records, a Suspicion Score is calculated.

Output Stage: A structured Risk Report is generated, highlighting interaction severity (Contraindicated, Major, Moderate) and providing an authenticity verdict.

3. Solution

We have built PharmaAI, a comprehensive web application that acts as a digital pharmacist.

Innovation: We moved away from hardcoded interaction tables. Our solution uses Live FDA Label Scraping, ensuring that if a drug warning is updated today, our users know today.

Practicality: By implementing a Two-Pass OCR, we solve the real-world problem of blurry or angled photos taken in low-light pharmacy environments.

Impact: It provides an immediate "Green/Yellow/Red" signal to the user, simplifying complex medical jargon into easy-to-understand safety alerts.

4. Technology Used

Python: The backbone of the entire pipeline logic.

OpenCV & PIL: Used for image preprocessing and enhancement to ensure the OCR engine gets the clearest possible text.

OpenRouter/Vision-Language Models: For high-accuracy extraction of structured data from unstructured medical packaging.

RESTful APIs (RxNorm, PubChem, OpenFDA): These serve as our "Single Source of Truth," providing clinical-grade data without the need for a local, heavy database.

JSON-based Reporting: Ensures that the backend can easily integrate with any modern web or mobile frontend.

5. Feature USP (Unique Selling Point)

Double-Verification OCR: Our "Two-Pass" system runs a second pass to cross-verify the first result, drastically reducing "hallucinations" in reading drug names.

The "Suspicion" Algorithm: Unlike other apps, we don't just identify the drug; we look for what's missing (like Batch/Exp dates) to flag potential counterfeits.

Real-time Risk Heatmap: Interactions are not just listed; they are ranked by clinical severity and displayed with visual emojis for instant recognition.

Zero-Database Dependency: By fetching data directly from the NLM and FDA, our solution is always up-to-date and highly scalable.

6. **Resources & Repositories**

- GitHub Repository: [Insert your Link Here]
- API Documentation: RxNorm REST API, OpenFDA.
- Reference Papers: Based on the "National Patient Safety Goals" for reducing harm from anticoagulation therapy and polypharmacy.

Conclusion: The Future of Digital Patient Safety

PharmaAI is more than just an OCR tool; it is a foundational step toward a transparent and safe pharmaceutical ecosystem. By bridging the gap between complex clinical data (FDA/RxNorm) and the everyday consumer, we have created a solution that is:

Scalable: Built on a "Zero-Database" architecture, the system stays updated automatically as global medical authorities update their records.

Preventative: By identifying both counterfeit markers and dangerous drug interactions simultaneously, we provide a proactive safety net rather than a reactive treatment.

Empowering: We provide patients and pharmacists with high-level medical intelligence—once reserved for clinical specialists—in the palm of their hands.

Final Vision: Our goal is to evolve PharmaAI into a global standard for medication verification, integrating with national health stacks like ABDM (Ayushman Bharat Digital Mission) to ensure that every pill taken is authentic, compatible, and safe.

Team Name:

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